1. Consider a memoryless binary symmetric channel. Show that for every codeword $\underline{c} \in \mathcal{C}$, the probability of error given that $\underline{c}$ is transmitted through the memoryless BSC is bounded as
$$P_e(\underline{c}) \leq \sum_{\underline{c}' \in \mathcal{C} \setminus \{\underline{c}\}} \left(\prod_{j=1}^{n} \sum_{y \in \{0,1\}} \sqrt{\text{Prob}(y|c_j)\text{Prob}(y|c_j')} \right),$$
where $\underline{c} = (c_1, c_2, \ldots, c_n)$ and $\underline{c}' = (c_1', c_2', \ldots, c_n')$.

2. The weight enumerator of a (n, M, d) code $\mathcal{C}$ is a polynomial in a variable z, denoted by $W_{\mathcal{C}}(z) = \sum_{i=0}^{n} A_i z^i$, where A_i is the number of codewords in $\mathcal{C}$ with Hamming weight i. Determine the values of $A_i, 0 \leq i \leq d-1$. Determine all the $\{A_i\}$ for the (7,4) Hamming code.

3. We have discussed Hamming code formed by three circles in the class. Describe a generalization of Hamming code by considering four circles (exactly similar to that of three circles). Determine the length of the code, size of the code, rate of the code and minimum distance of the code.

4. Suppose that we are given n-length binary vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$, such that the following relation holds, for a fixed $t > 0$:
$$d_H(\mathbf{u}, \mathbf{v}) = d_H(\mathbf{v}, \mathbf{w}) = d_H(\mathbf{u}, \mathbf{w}) = 2t.$$
Construct a binary vector, $\mathbf{x}$, such that $d_H(\mathbf{u}, \mathbf{x}) = d_H(\mathbf{v}, \mathbf{x}) = d_H(\mathbf{w}, \mathbf{x}) = t$.

5. Let S denote the memoryless binary erasure channel with input alphabet $\mathbb{F}_2 = \{0, 1\}$, output alphabet $\Phi = \mathbb{F}_2 \cup \{?\}$ and erasure probability $p = 0.1$. A codeword of the binary $(n, M, d) = (4, 8, 2)$ simple parity check code is transmitted through S and the following decoder $\mathcal{D} : \Phi^4 \to \mathcal{C} \cup \{''e''\}$ is applied to the received codeword:
$$\mathcal{D}(y) = \begin{cases} \underline{c} & \text{if } \underline{y} \text{ agrees with exactly one } \underline{c} \in \mathcal{C} \text{ on the unerased entries} \\ ''e'' & \text{otherwise} \end{cases}$$
Compute the probability that $\mathcal{D}$ produced $''e''$. Does this probability depend on which codeword is transmitted?

6. Let $\mathcal{C}$ be an (n, M, d) code over $\mathbb{F}_2$ and that τ and σ be non-negative integers such that $2\tau + \sigma \leq d - 1$. Show that there is a decoder $\mathcal{D} : \mathbb{F}_2^n \to \mathcal{C} \cup \{''e''\}$ with the following properties:
   - If number of errors is τ or less, then errors will be recovered correctly.
   - Otherwise if the number of errors is $\tau + \sigma$ or less, then they will be detected.

7. Consider an (n, k) linear block code. For any set of k independent columns of the generator matrix G, the corresponding set of coordinates $S \subseteq \{1, 2, \ldots, n\}$ is defined as an information set. Determine five information sets of the (7,4) Hamming code discussed in the class.